

Common Ports You Should Know

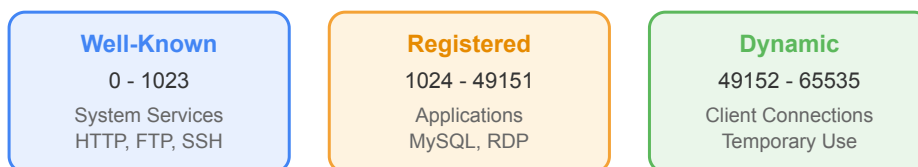
In computer networking, **ports** serve as logical channels that enable communication between devices and applications. Each port is identified by a unique **port number** ranging from 0 to 65535, assigned by the Internet Assigned Numbers Authority (IANA). Understanding these port numbers and their associated services is essential for network administrators, IT professionals, and cybersecurity specialists.

Port Number Categories

Port numbers are divided into three distinct ranges, each serving different purposes:

- **Well-Known Ports (0-1023)** Reserved for system services and widely-used protocols
- **Registered Ports (1024-49151)** Assigned to user processes and specific applications upon request
- **Dynamic/Private Ports (49152-65535)** Used by client applications for temporary connections

Port Number Ranges



Total: 65,536 possible port numbers (0-65535)

Essential Well-Known Ports (0-1023)

These ports are reserved for critical system services and standard protocols:

- **Port 21** FTP (File Transfer Protocol) for file transfers
- **Port 22** SSH (Secure Shell) for secure remote access
- **Port 23** Telnet for remote terminal access (insecure)
- **Port 25** SMTP (Simple Mail Transfer Protocol) for email sending
- **Port 53** DNS (Domain Name System) for name resolution
- **Port 67/68** DHCP (Dynamic Host Configuration Protocol) for IP assignment
- **Port 80** HTTP (HyperText Transfer Protocol) for web traffic

- **Port 110** POP3 (Post Office Protocol v3) for email retrieval
- **Port 123** NTP (Network Time Protocol) for time synchronization
- **Port 143** IMAP (Internet Message Access Protocol) for email access
- **Port 161/162** SNMP (Simple Network Management Protocol) for network monitoring
- **Port 389/636** LDAP/LDAPS (Lightweight Directory Access Protocol)
- **Port 443** HTTPS (HTTP Secure) for encrypted web traffic
- **Port 445** SMB (Server Message Block) for file and printer sharing
- **Port 993/995** IMAPS/POP3S for secure email protocols

Important Registered Ports (1024-49151)

These ports are assigned to specific applications and services:

- **Port 1080** SOCKS proxy server
- **Port 1433** Microsoft SQL Server database
- **Port 1521** Oracle database server
- **Port 1701** L2TP (Layer 2 Tunneling Protocol) for VPN
- **Port 1723** PPTP (Point-to-Point Tunneling Protocol) for VPN
- **Port 2049** NFS (Network File System) for file sharing
- **Port 3128** Squid proxy server
- **Port 3306** MySQL database server
- **Port 3389** RDP (Remote Desktop Protocol) for Windows remote access
- **Port 5060/5061** SIP (Session Initiation Protocol) for VoIP
- **Port 5222/5223** XMPP (Extensible Messaging and Presence Protocol)
- **Port 5353** mDNS (Multicast DNS) and Bonjour
- **Port 5432** PostgreSQL database server
- **Port 5900** VNC (Virtual Network Computing) for remote access
- **Port 6379** Redis key-value database
- **Port 6667/6697** IRC (Internet Relay Chat)

- **Port 8080** HTTP alternative and proxy servers
- **Port 11211** Memcached distributed caching system
- **Port 27017** MongoDB database server

Specialized and Additional Ports

- **Port 137-139** NetBIOS for Windows network communication
- **Port 500** IKE (Internet Key Exchange) for VPN key management
- **Port 548** AFP (Apple Filing Protocol) for Mac file sharing
- **Port 873** Rsync for remote file synchronization
- **Port 5355** LLMNR (Link-Local Multicast Name Resolution)
- **Port 5672** AMQP (Advanced Message Queuing Protocol)
- **Port 16384-32767** RTP/RTCP (Real-time Transport Protocol) for media streaming

Security and Best Practices

Understanding port numbers is crucial for network security. Misconfigured or unnecessarily open ports can create vulnerabilities and potential entry points for cyberattacks. Network administrators should regularly audit open ports, implement proper firewall rules, and follow the principle of least privilege when configuring network access.

Common Use Cases

- **Network troubleshooting** Identifying communication issues between services
- **Security auditing** Scanning for open ports and potential vulnerabilities
- **Firewall configuration** Creating rules to allow or block specific services
- **Service monitoring** Ensuring critical applications are accessible on their designated ports

Conclusion

These port numbers serve as the foundation for network communication, enabling everything from web browsing to database connections. Understanding port categories, their associated services, and security implications helps network professionals maintain robust, secure, and efficient network infrastructures.